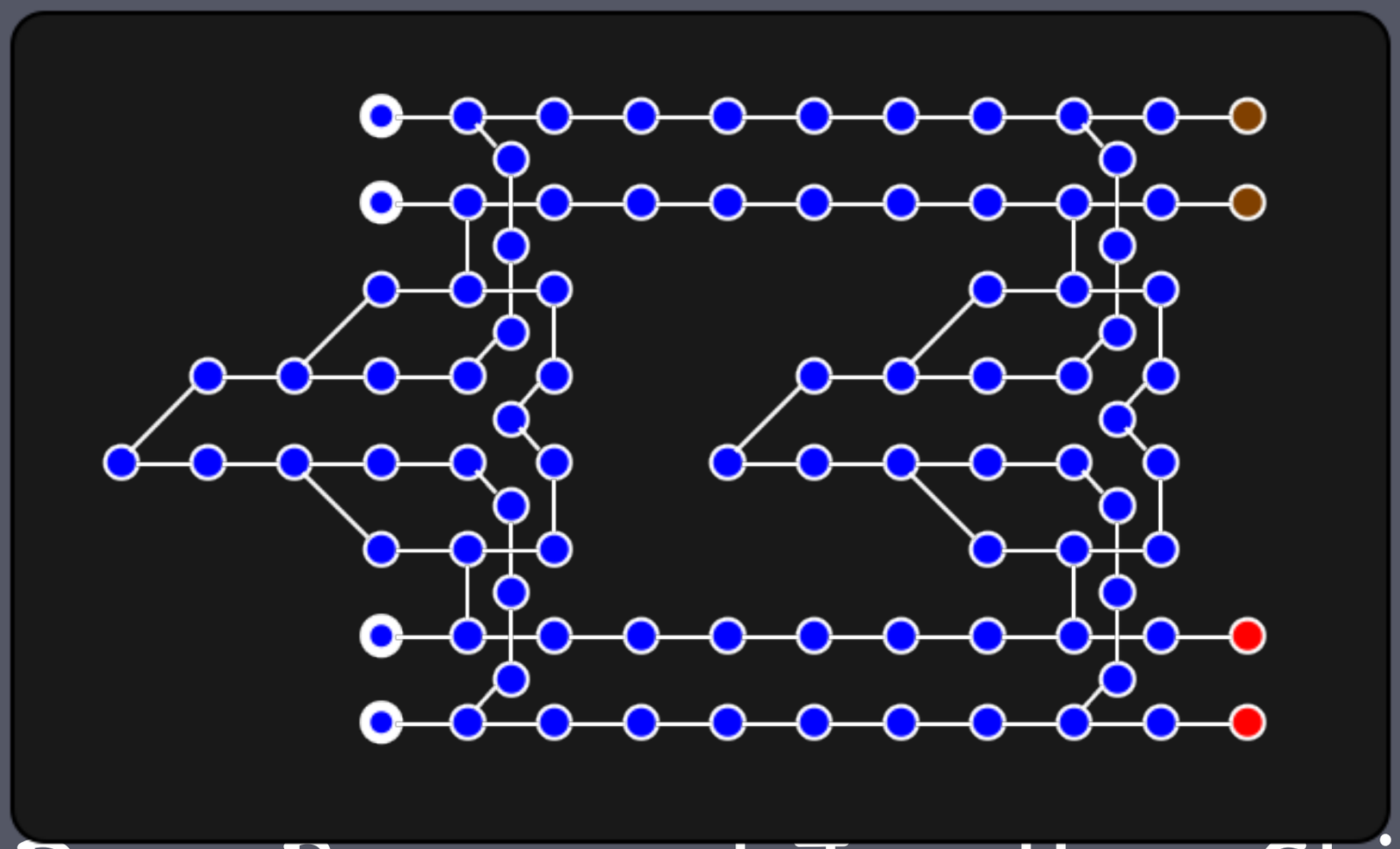
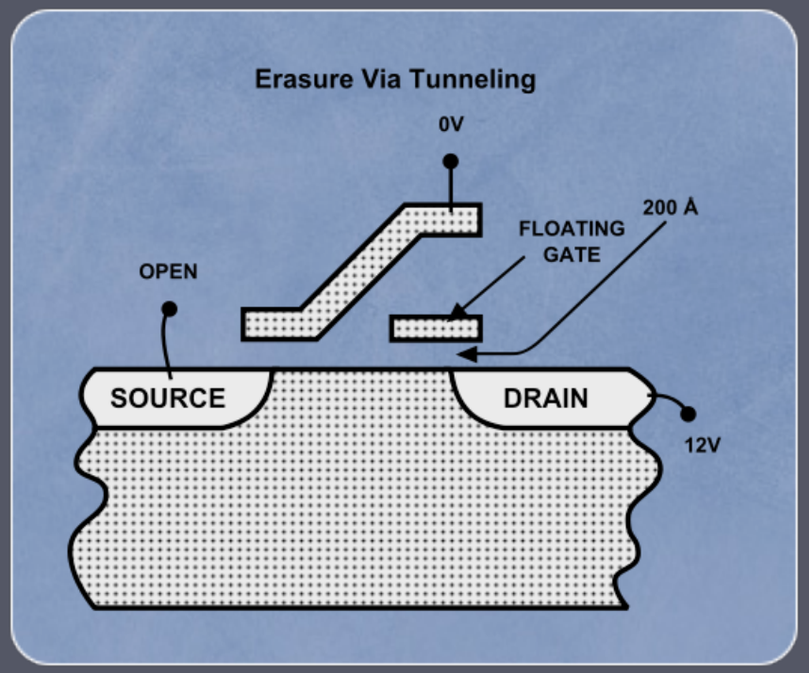
Making Error Correction Spatially Local

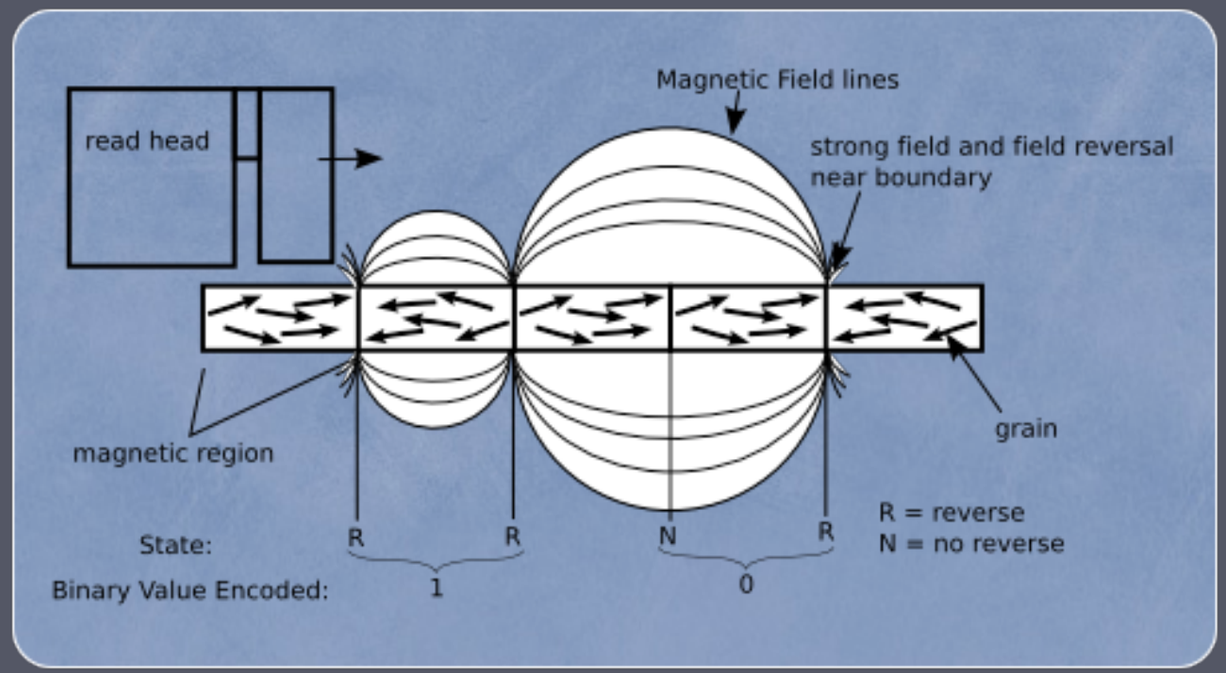


Dave Bacon and Jonathan Shi
University of Washington

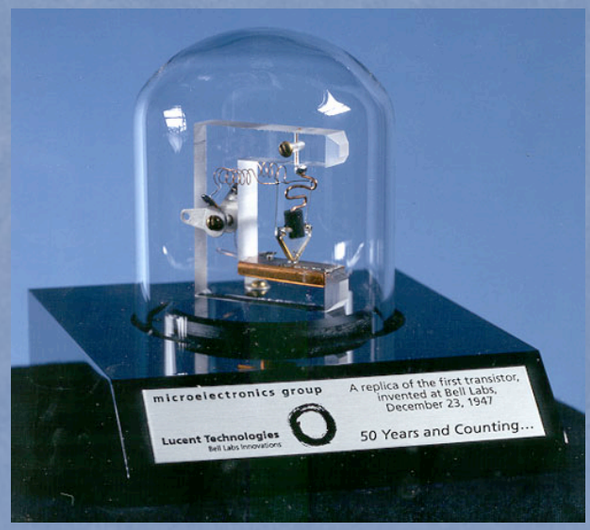
Error Correction



flash memory



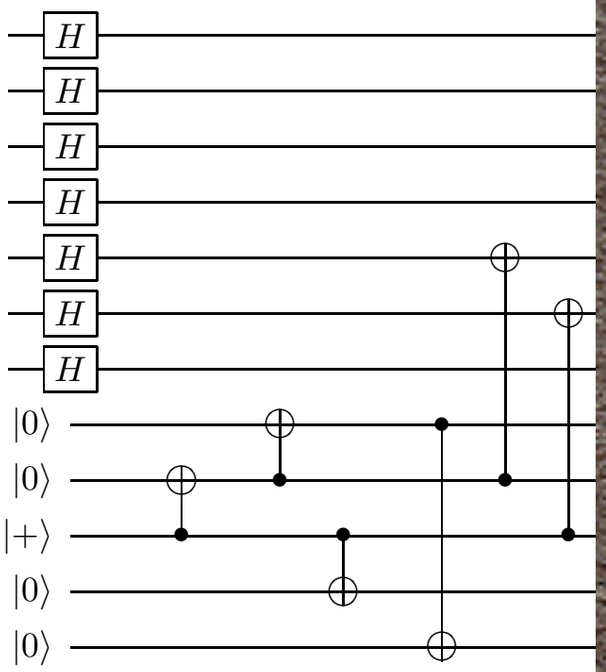
magnetic hard drive



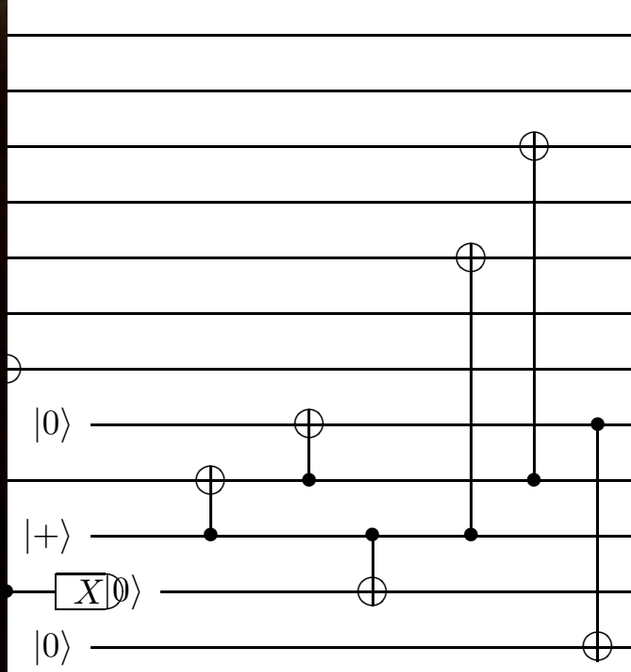
transistor

+ fast
+ cool
+ less
+ uses
- subject
- wears
- requires
- error correction

Quar ection





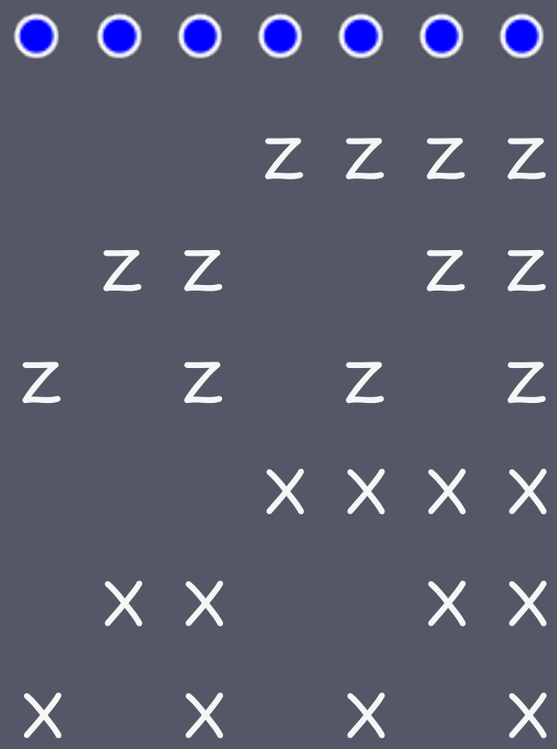


Compl be for
pr on

I CANZ FIT N XPERIMENTALST'S BOX?

Stabilizer Codes

[[7,1,3]] Steane code

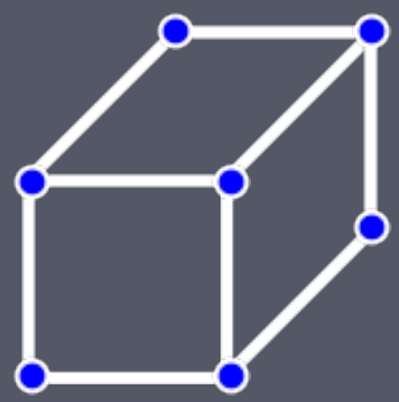


row = stabilizer

column = qubit

QEC Recipe

1. Measure 6 stabilizer operators
2. Conditional on measurement diagnose (and fix) single qubit error



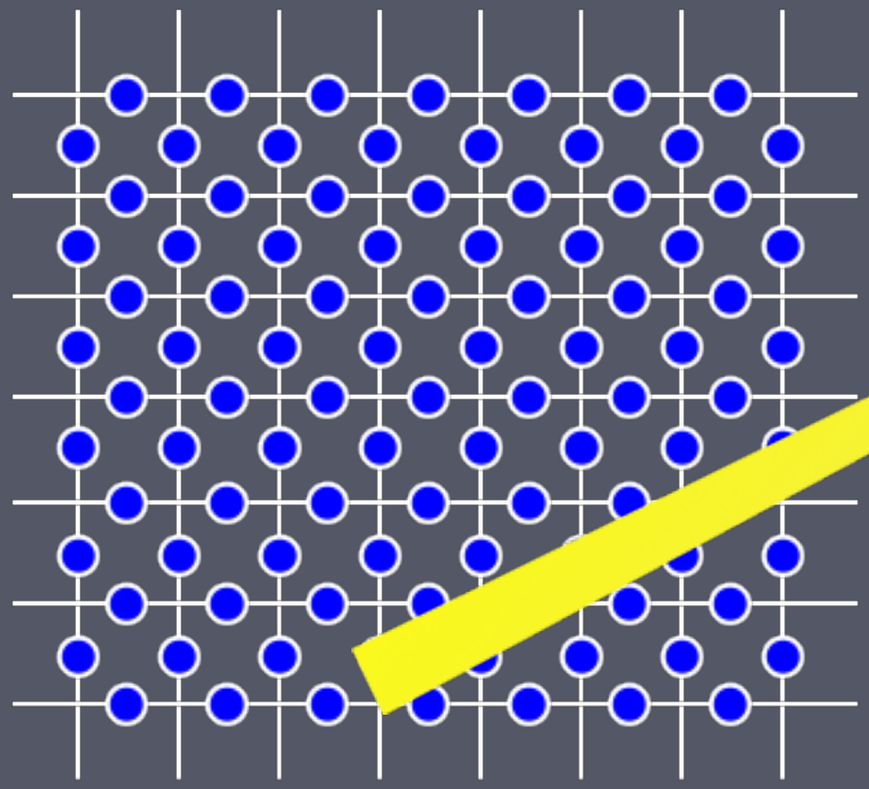
[[# qubits, # encoded qubits, distance]]

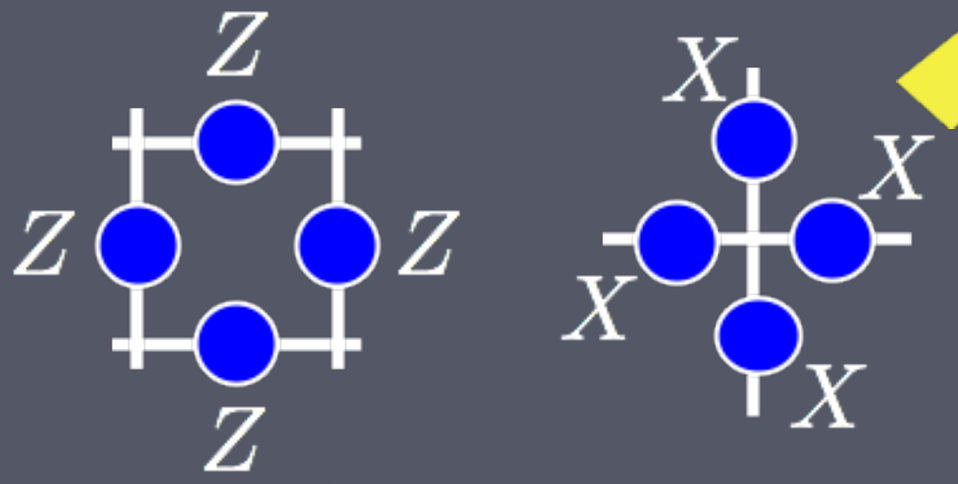
Concat es Away



Concatena f spatially distant, hig

Rays of Hope





plaquette vertex

Surface code QEC

1. Measure plaquette and vertex operators

2. Classically post-process measurement results to diagnose error

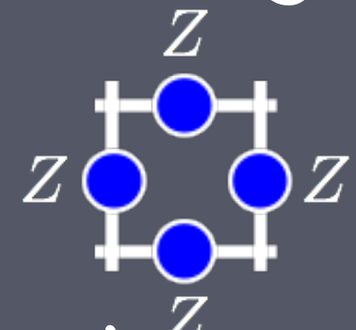
k-Local and k-Neighboring

weight of Pauli operator = # of operators ≠ I

➡ $I^{\otimes}X^{\otimes}X^{\otimes}Y^{\otimes}Y^{\otimes}I^{\otimes}Z^{\otimes}I$ has weight 5

k-Local code: exists a set of stabilizer generators whose weight ≤ k

➡ surface code is 4-local



k-Neighboring code (in D dimensions): exists a set of stabilizer generators where all ≠ I operators are within a distance k of each other on some lattice of qubits in D spatial dimensions

➡ surface code is 2-neighboring in 2D

Lament



Why can't all codes be k-neighboring in two or three spatial dimensions?

Main Result

For every [[n,k,d]] stabilizer code, there exists a [[N,k,r,d]] st… …e, $N=O(n^2)$, whose syndr… …n be made using k-neig… …s (k constant).

Every st… …t

…!!



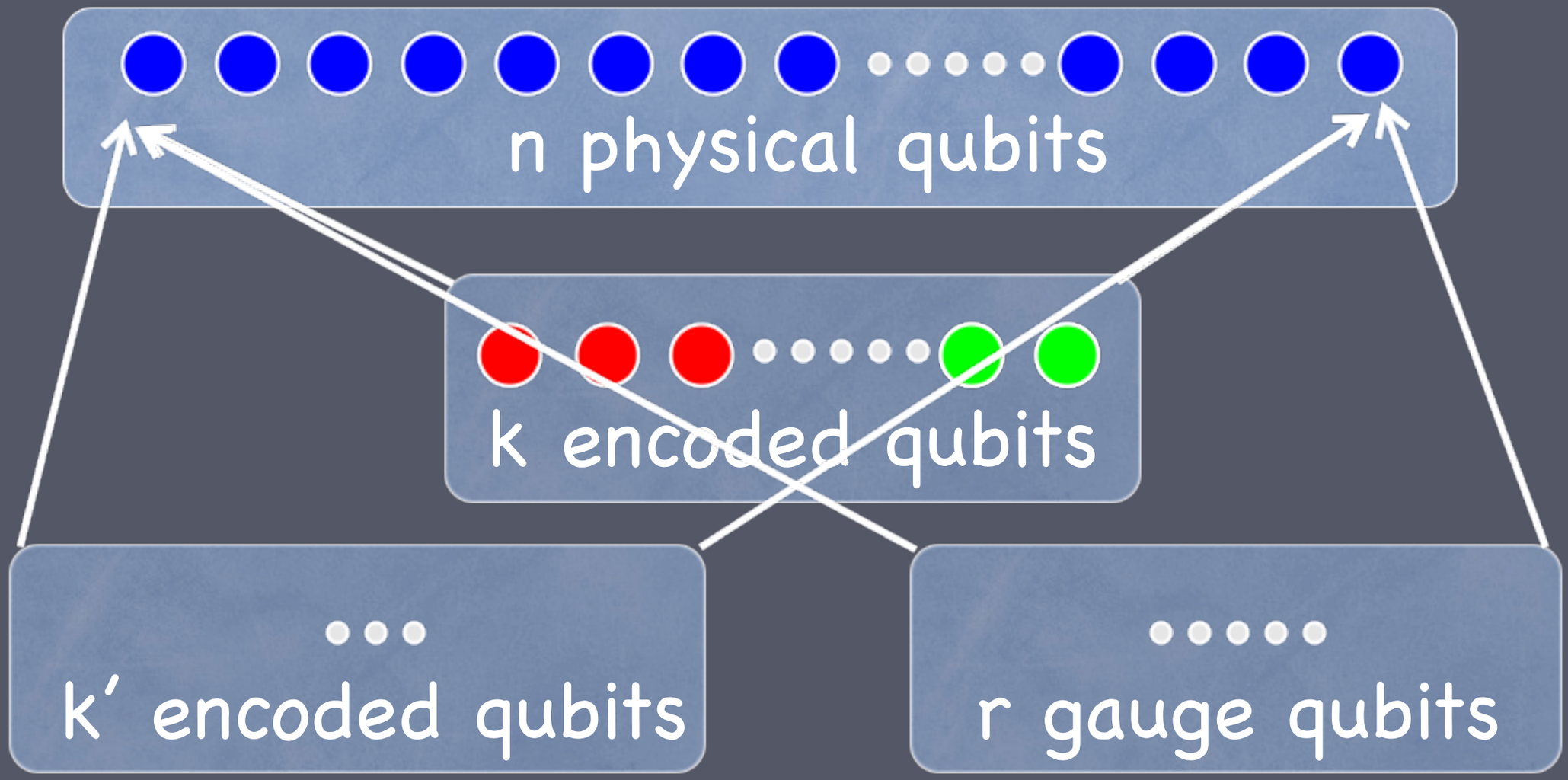
Main Result

For every [[n,k,d]] stabilizer code, there exists a [[N,k,r,d]] stabilizer subsystem code, $N=O(n^2)$, whose syndrome measurements can be made using k-neighboring measurements (k constant).

1. stabilizer subsystem codes
2. spaced-time locality in QEC
3. measurement based quantum computing

Stabilizer Subsystem Code

[n,k,d] Stabilizer Subspace Code

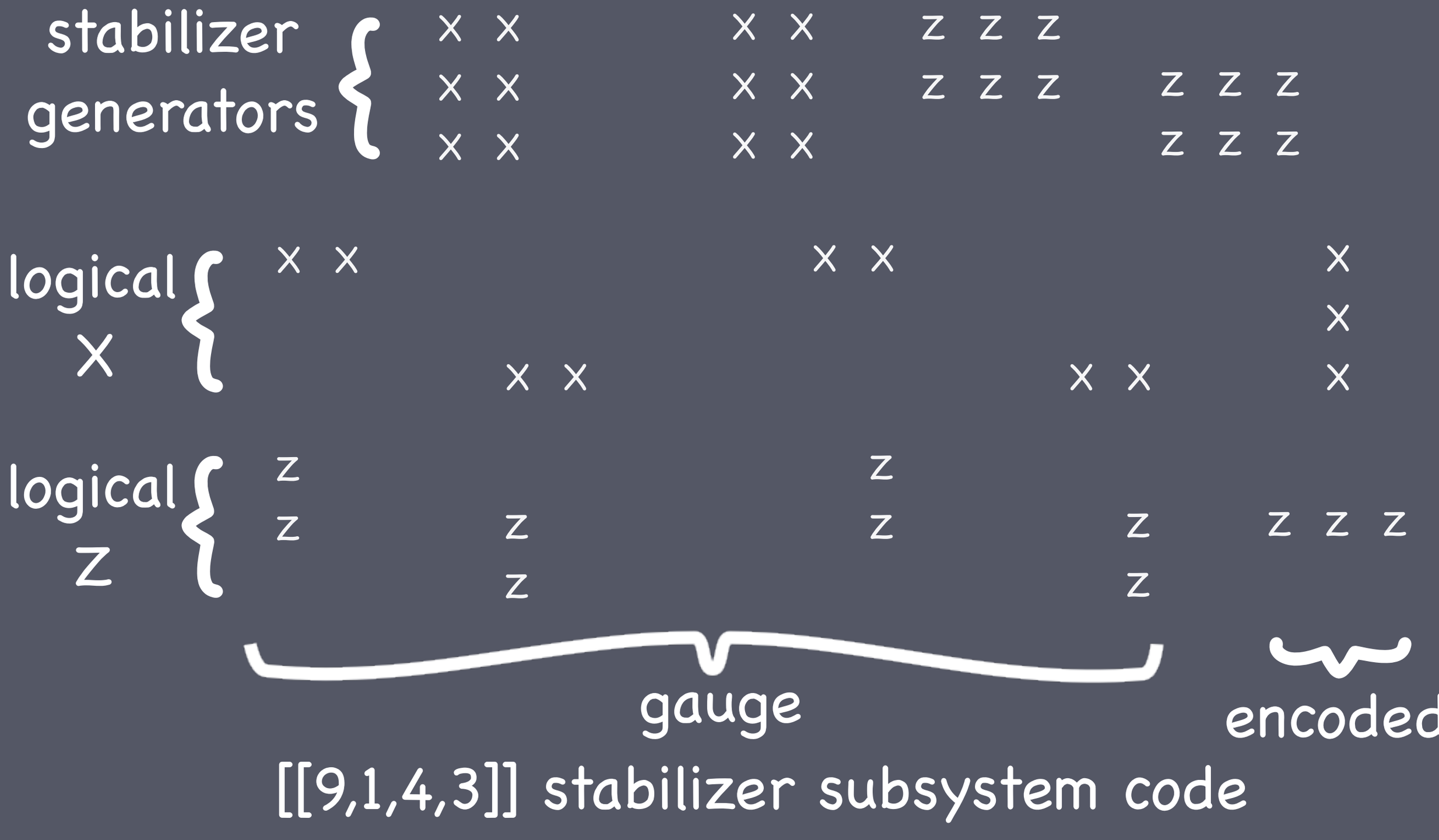


[n,k′,r,d] Stabilizer Subsystem Code

[David Poulin, arXiv:quant-ph/0508131]

Baloney Sandwich Code

[Bacon, arXiv:quant-ph/0506023]



[[9,1,4,3]] stabilizer subsystem code

Baloney Sandwich Code

stabilizer generators {



S_1



S_2



S_3



S_4



=





S_1 X_1 X_2 $X_1X_2S_1$

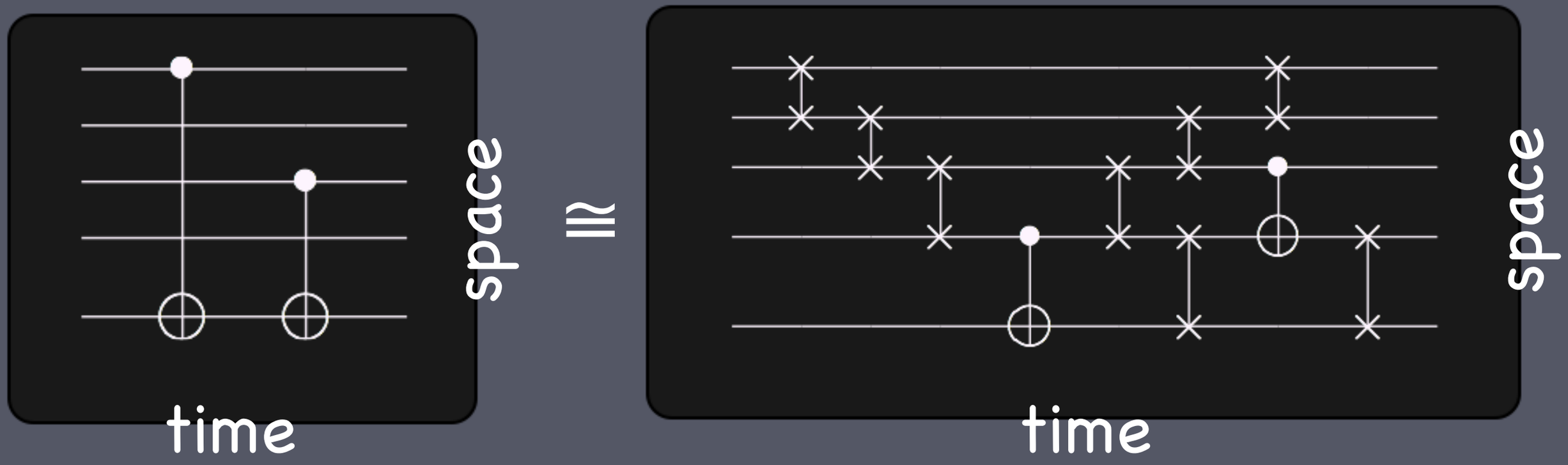
We can measure S_1 by measuring 2-local operators that are products of gauge and stabilizer operators

6-local → 2-local

Space-Time Neighboring

Any quantum circuit can be made to be neighboring in space-time for any spatial dimension D

Example in 1D



Space-Time QEC

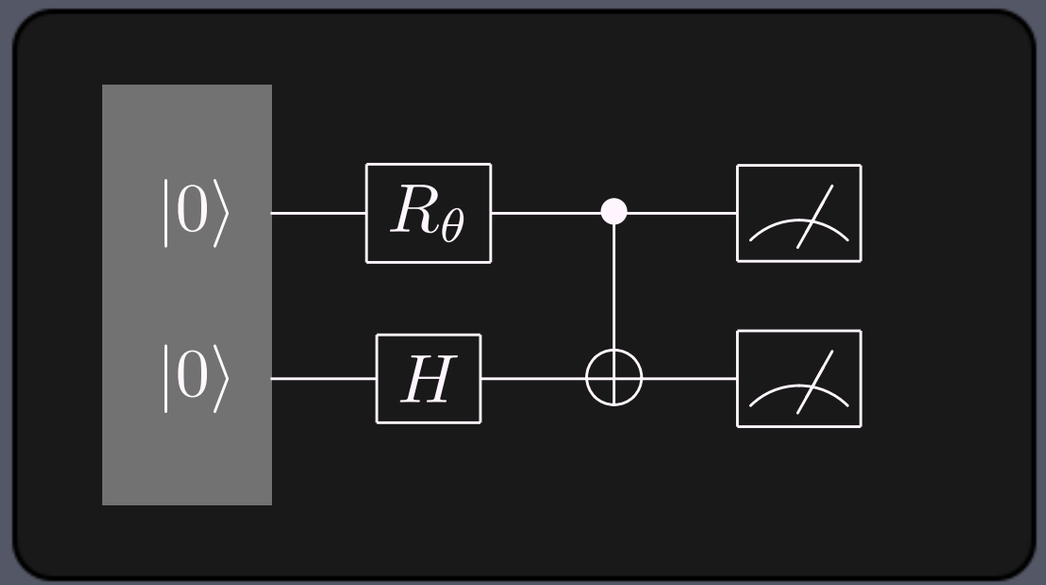
Fault-tolerant quantum error correction can be done using k-neighboring locality in space-time while maintaining a threshold.

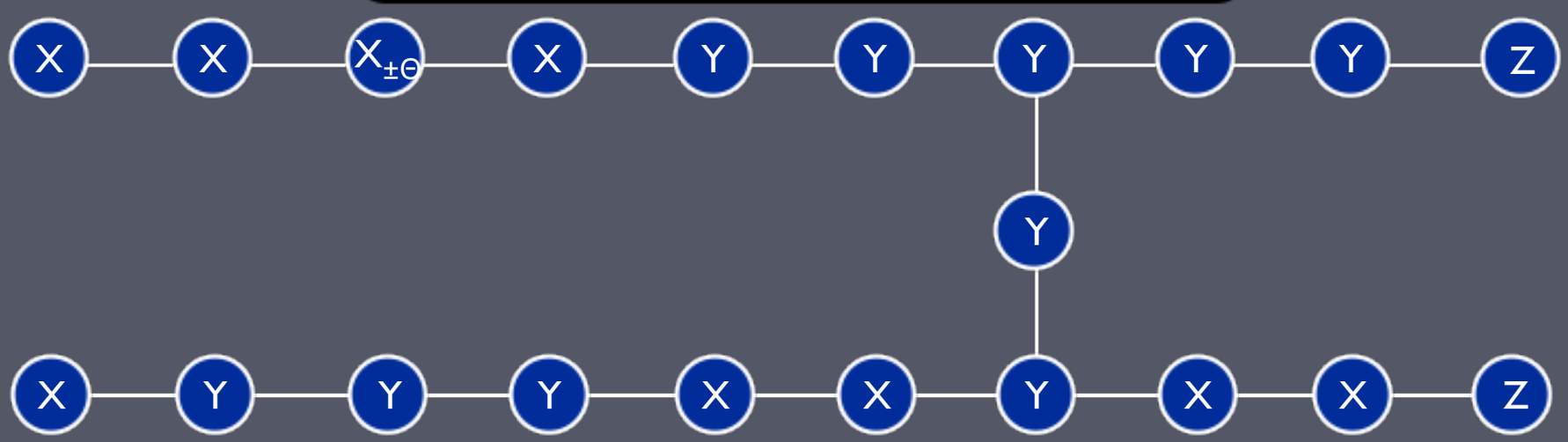
(1D requires next-nearest neighbor gates)

[Aharonov, Ben-Or, arXiv:quant-ph/9906129]
[Gottesman, arXiv:quant-ph/9903099]

Measurement Based QC

[Raussendorf and Briegel, Phys. Rev. Lett. 86, 5188 (2001)]

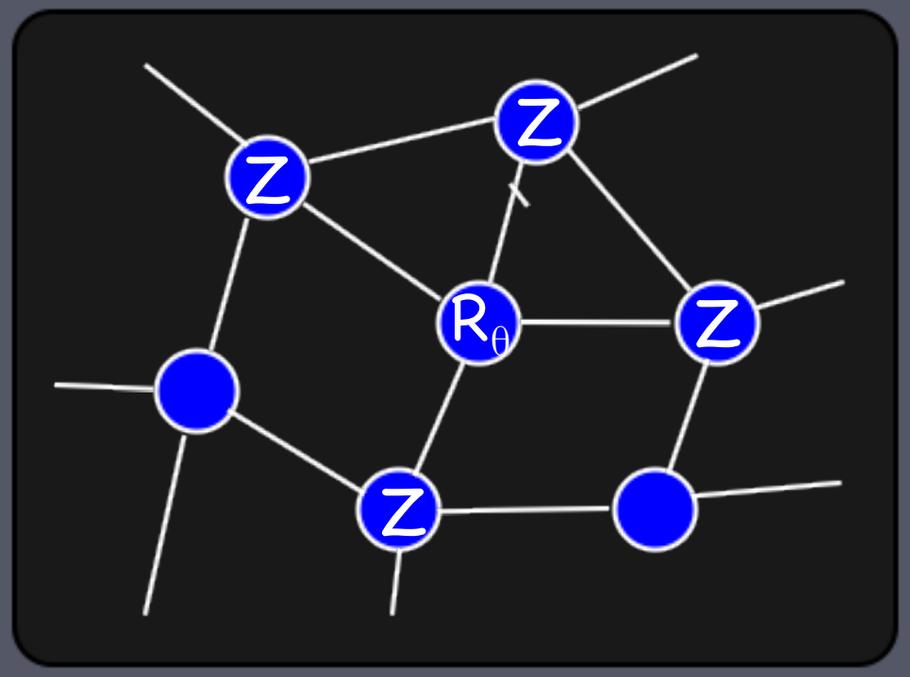




Adaptive Create entangled state Entangling gates Measurements Local circuit

Initial state described locally + measurements are all local

Twisted Graph State



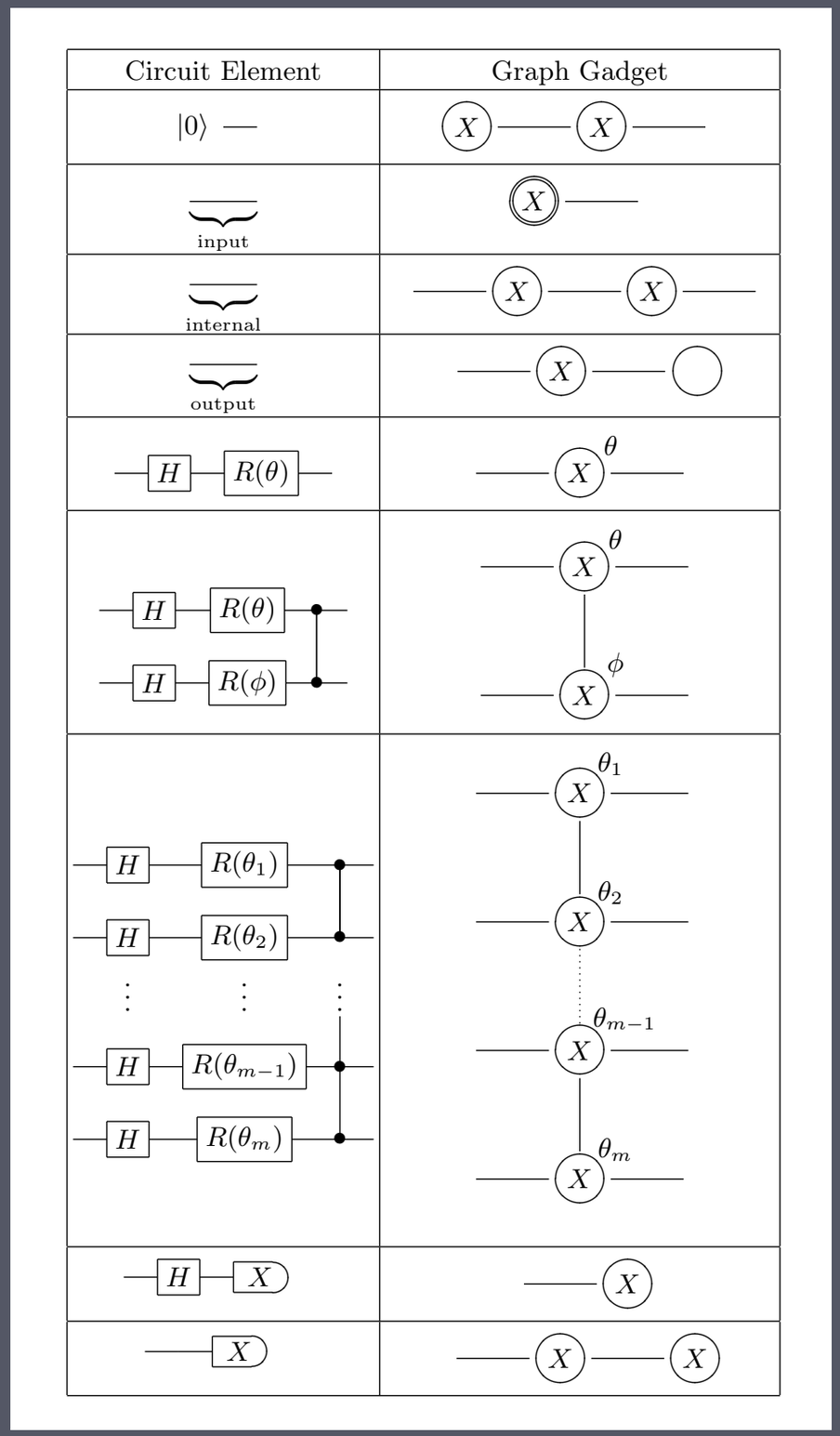
$$= \quad [R_\theta]_v \quad \prod_{w|(v,w)\in\mathcal{E}} \quad [Z]_w$$

$$R_\theta = \cos\theta X + \sin\theta Y$$

"Vertex operator with angle θ"

Twisted Graph State: is common +1 eigenstate of all vertex operators (one per vertex)

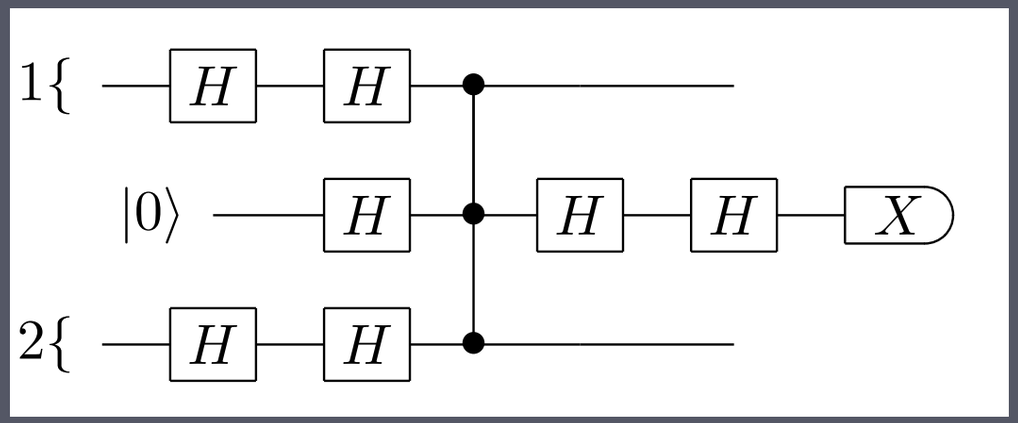
QC to Twist Graph State

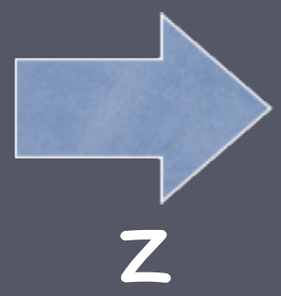


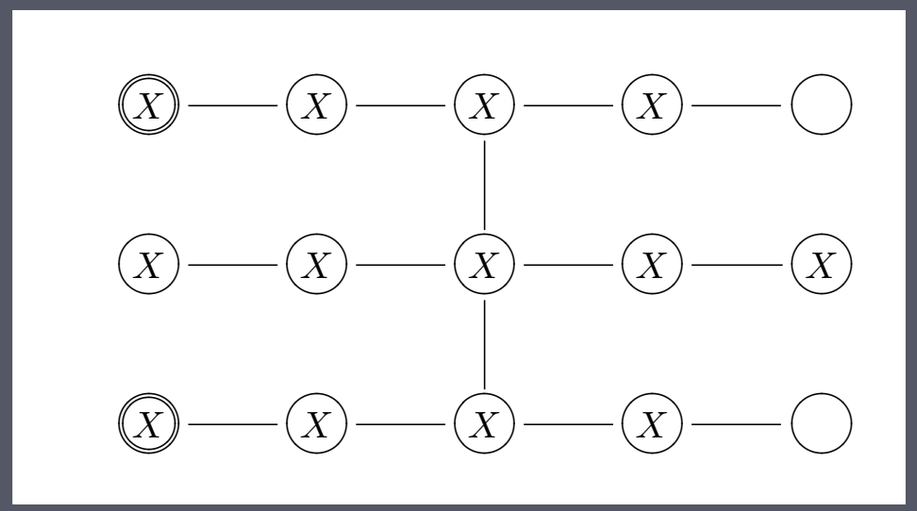
Quantum circuit made up of preparations, gates, and measurements

Graph describing vertex operators and places where X measurements would be made to enact MBQC for the quantum circuit

Encoded MBQC Example





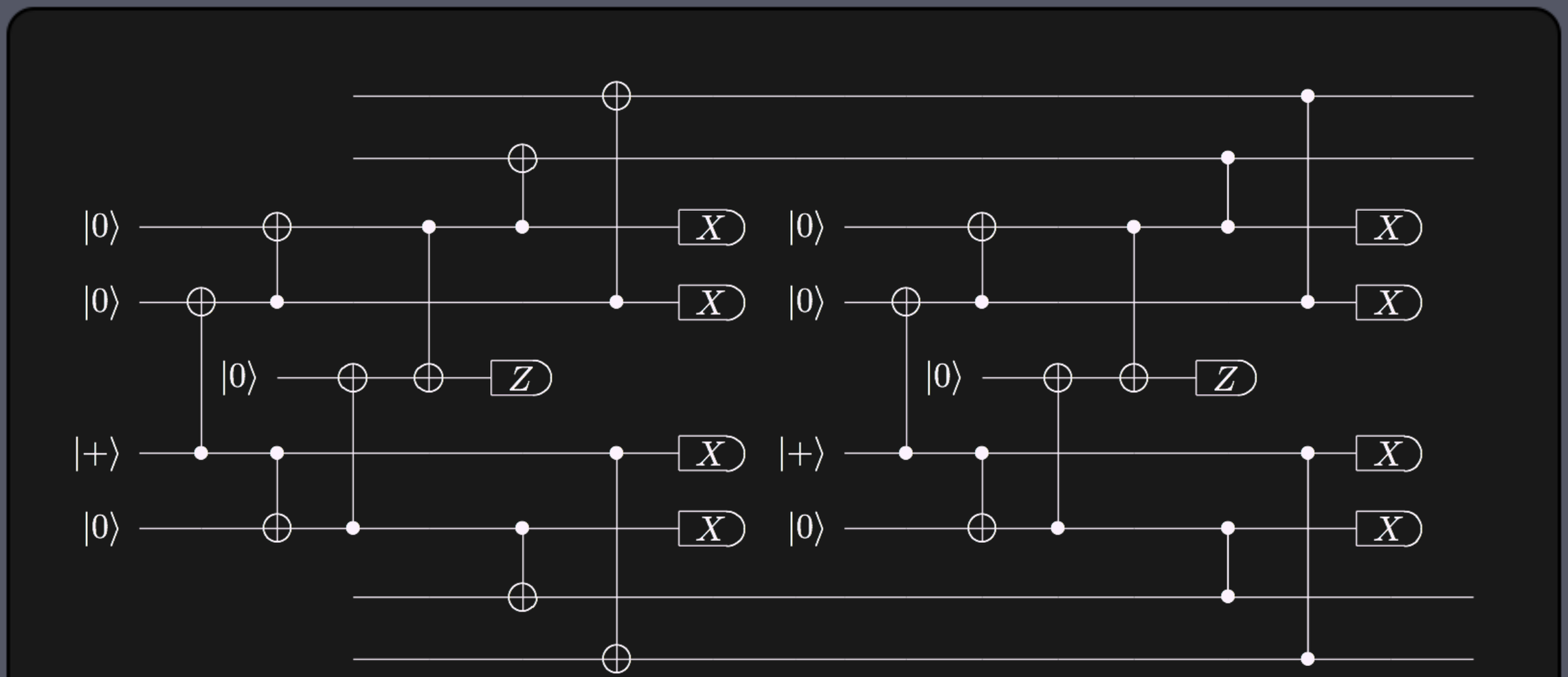


1. vertex operator for every non-double-circled node
2. starting state is +1 common eigenspace of these vertex operators
3. input information encoded @ double-circled nodes
4. X measurements occur on all vertices labeled X
5. Output measurements at X's, unmeasured output qubits at non-X-ed nodes

CONSTRUCTION

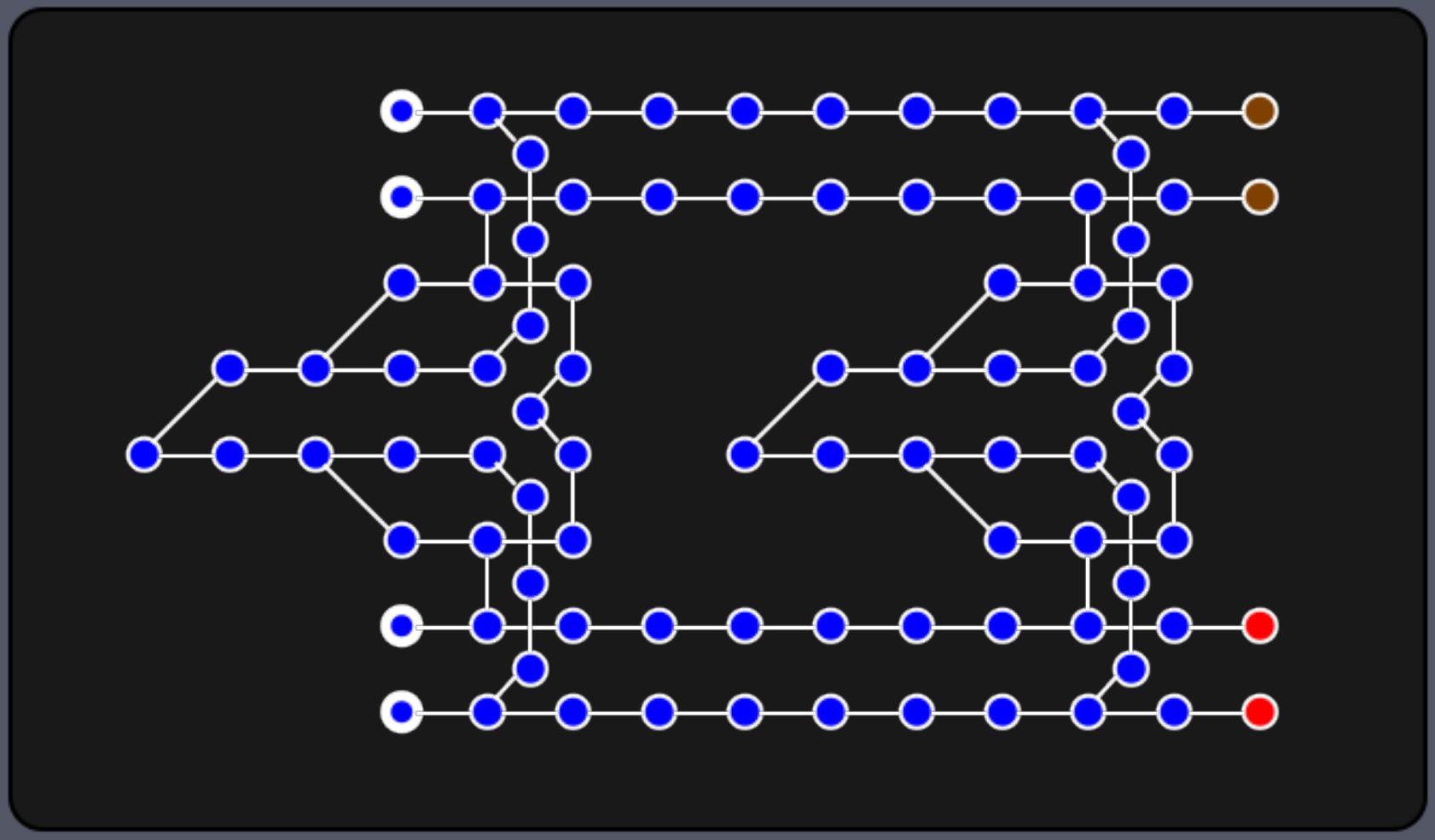
Input: [[n,k,d]] stabilizer code

1. Construct fault-tolerant syndrome measuring quantum circuit that is k-neighboring in space-time on lattice in dimension D (D>1)



CONSTRUCTION

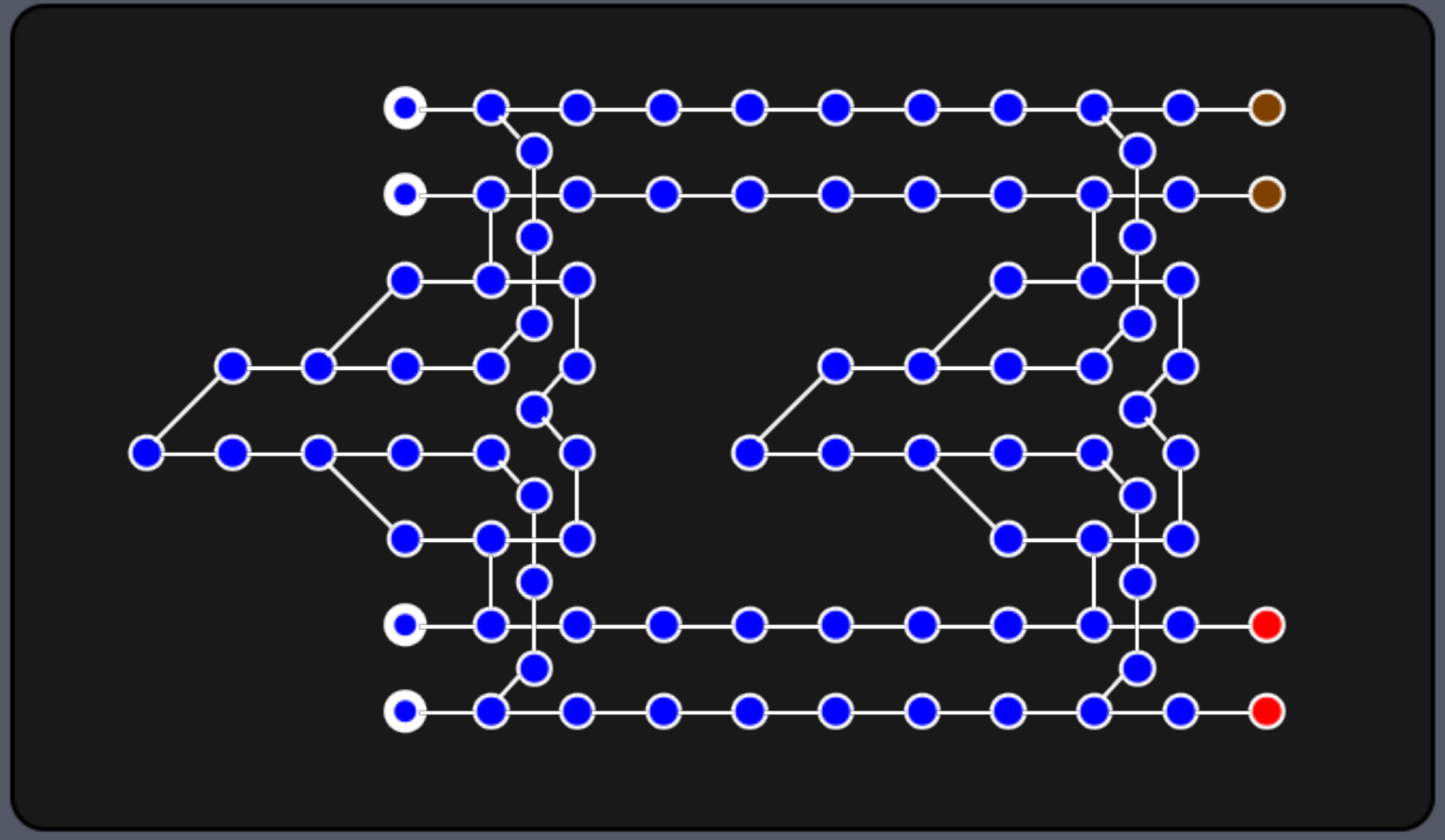
2. Convert this circuit to the scheme that would be used to implement it in MBQC



[Related: Rausendorf, Harrington, Goyal 2006]

CONSTRUCTION

3. Take operators from MBQC construction (vertex operators and X operators for measurements) and identify (a) stabilizer subgroup of these operators and (b) gauge qubits

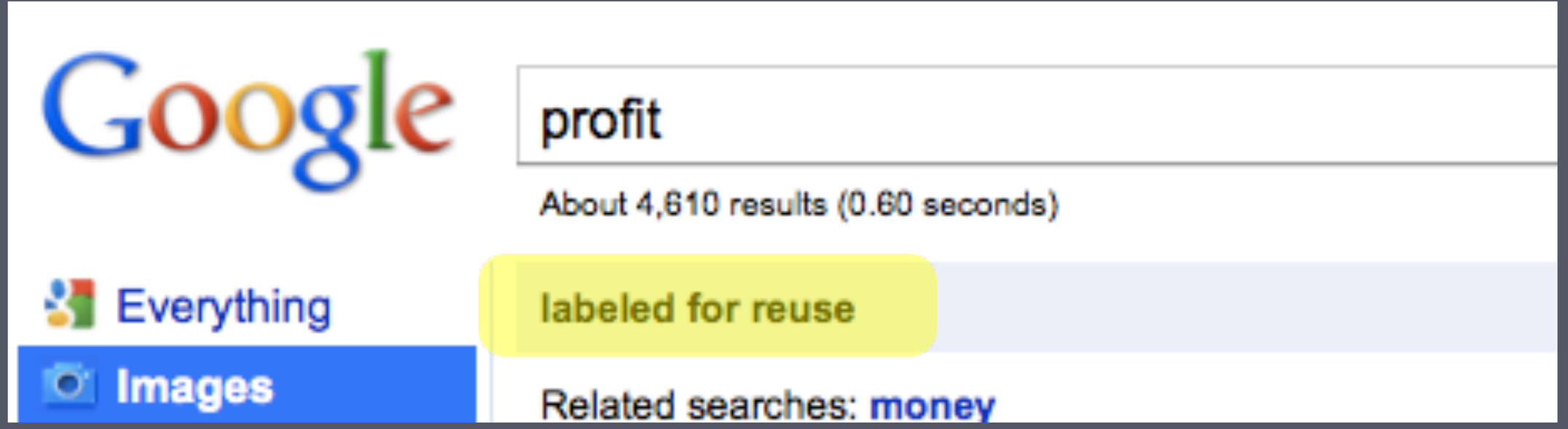


$X^{\otimes 4}$ and $Z^{\otimes 4}$ stabilizer localized on red qubits

Profit

For every [[n,k,d]] stabilizer code, there exists a [[N,k,r,d]] stabilizer subsystem code, $N=O(n^2)$, whose syndrome measurements can be made using k-neighboring measurements (k constant).

$$$$



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profit

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Photo
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Line drawing

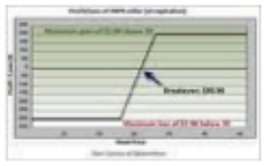
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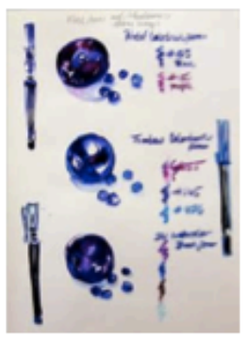
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Related searches: money



















FAQ

Q: Why fault-tolerant syndrome measuring circuits?
- These maintain the distance of the code
Q: But aren't fault-tolerant circuits probabilistic?
- Often yes. Use the circuit as if the measurement result turned out in your favor
Q: Do these codes have thresholds?
- Unknown, but likely given their relation to fault-tolerant circuits
Q: k-local for what k?
- Can be made 3-local

Consequences



arXiv:soon

Q-Dub Group

<u>Grads</u>

Greg Crosswhite (Physics)
Isaac Crosson (Physics)
Lukas Svec (Physics)
Jijiang Yan (Physics)
Kamil Michnicki (Physics)
David Rosenbaum (CS)
Paul Pham (CS)
Kevin Zatloukal (CS)
Punya Biswall (CS)

<u>Faculty</u>

Dave Bacon (Pseudo-prof)
Aram Harrow (Pseudo-prof)

<u>Undergrads</u>

Zakk Webb (Physics/CS/Math)
Kate Liotta (Physics/CS)
Jonathan Shi (Physics/CS)
Melanie Jensenworth (Math/CS)
Jeffrey Booth (CS)
Rob McClure (CS)
Harshad Petwe (CS)

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The Stubbornest Dog

